

Grade 8
Unit 12
Lesson 8 Practice

Name _____
Date _____
Period _____

For questions 1-3, consider the following context.

Romeo is evaluating the expression $-5(-\sqrt{100} + 25)$. Instead of first performing operations within the parentheses, he multiplied the -5 by both terms in the parentheses, as shown.

$$-5(-\sqrt{100} + 25) = (-5 \cdot -\sqrt{100}) + (-5 \cdot 25)$$

1. Which property of operations did Romeo use?
2. Evaluate both expressions that Romeo set equal to each other using the order of operations.
3. Explain how the values of the two expressions compare.

For questions 4-5, consider the following context.

Cynthia says that when following the order of operations, the two expressions below are equivalent and the parentheses are not needed.

$$(-2 \cdot \sqrt{64}) + 17 = -2 \cdot \sqrt{64} + 17$$

4. Explain whether Cynthia is correct. Justify your answer mathematically.
5. Would the parentheses change the value of the expression if they were around $\sqrt{64} + 17$ instead? Justify your answer mathematically.



For questions 6-10, find the value of each expression.

6. $\left(\frac{1}{7} \cdot 7 - \frac{1}{4}\right) \cdot \sqrt{16}^3$

7. $\sqrt{\frac{9^2-9}{2}} - 4.5 + \frac{1}{4}$

8. $\frac{1}{4} \cdot \left(7 - \frac{1}{3}\sqrt{225}\right)^3$

9. $\sqrt[3]{-125} - 7 \cdot 2^2 + \left(\frac{1}{9}\right)^{-1}$

10. $\left(\frac{1}{4} \cdot 16\right)^{-2} - \frac{5}{8} \cdot 24$